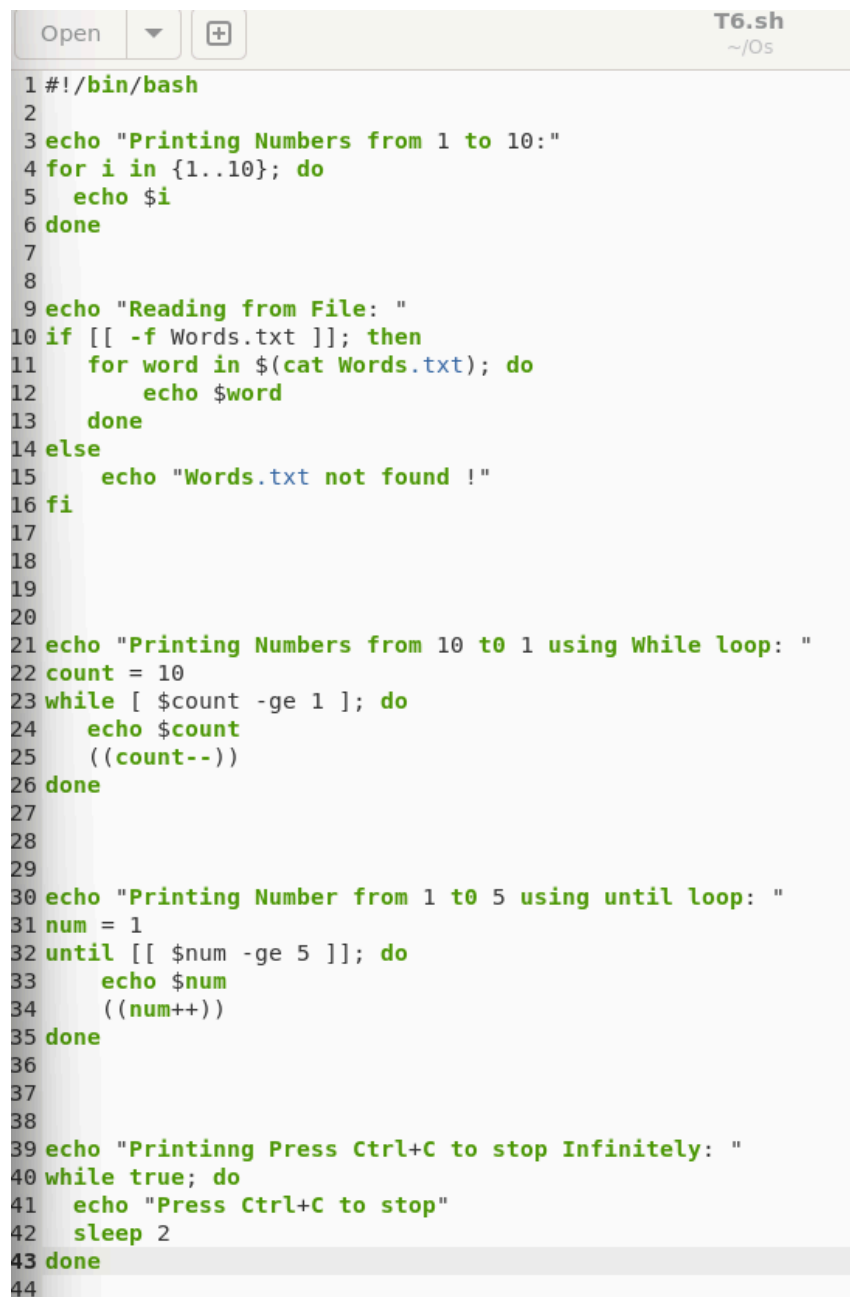


OS Lab 03(B) Submission

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Roll Number: 24k-0737

Task 06:



```
1 #!/bin/bash
2
3 echo "Printing Numbers from 1 to 10:"
4 for i in {1..10}; do
5     echo $i
6 done
7
8
9 echo "Reading from File: "
10 if [[ -f Words.txt ]]; then
11     for word in $(cat Words.txt); do
12         echo $word
13     done
14 else
15     echo "Words.txt not found !"
16 fi
17
18
19
20
21 echo "Printing Numbers from 10 to 1 using While loop: "
22 count = 10
23 while [ $count -ge 1 ]; do
24     echo $count
25     ((count--))
26 done
27
28
29
30 echo "Printing Number from 1 to 5 using until loop: "
31 num = 1
32 until [[ $num -ge 5 ]]; do
33     echo $num
34     ((num++))
35 done
36
37
38
39 echo "Printinng Press Ctrl+C to stop Infinitely: "
40 while true; do
41     echo "Press Ctrl+C to stop"
42     sleep 2
43 done
44
```

Output:

```
yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh
Printing Numbers from 1 to 10:
1
2
3
4
5
6
7
8
9
10
```

```
Reading from File:
Hello
My
name
is
yash
raj.
I
am
a
Fastian
!
My
Lab
Instructor
name
is
:
Sir
Abdullah
```

```

Adductan
Printing Numbers from 10 to 1 using While loop:
10
9
8
7
6
5
4
3
2
1
Printing Number from 1 to 5 using until loop:
1
2
3
4
Printinng Press Ctrl+C to stop Infinitely:
Press Ctrl+C to stop
Press Ctrl+C to stop

```

Task 07:

```

1 #!/bin/bash
2
3 file="Words.txt" # Pass the filename as argument
4 if [[ ! -f $file ]]; then
5     echo "File not found!"
6     exit 1
7 fi
8
9 count=0
10
11 while IFS= read -r line; do
12     if [[ $line == *"Linux"* ]]; then
13         echo "Found Linux: $line"
14         ((count++))
15     fi
16 done < "$file"
17
18 echo "Total lines containing 'Linux': $count"

```

Output:

```

yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh
Total lines containing 'Linux': 0
yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh

```

Task 08:

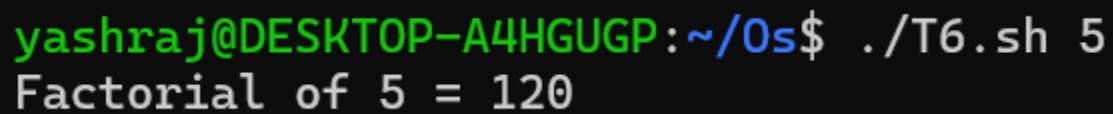
#!/bin/bash

```
factorial() {  
    local n=$1  
    local result=1  
    for (( i=2; i<=n; i++ )); do  
        ((result *= i))  
    done  
    echo $result  
}
```

```
if [[ $# -lt 1 ]]; then  
    echo "Usage: $0 number [number2 number3 ...]"  
    exit 1  
fi
```

```
for num in "$@"; do  
    echo "Factorial of $num = $(factorial $num)"  
done
```

Output:

A terminal window with a black background. The prompt is 'yashraj@DESKTOP-A4HGUGP:~/Os\$' in green and blue. The command './T6.sh 5' is entered in white. The output 'Factorial of 5 = 120' is displayed in white on the next line.

```
yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh 5  
Factorial of 5 = 120
```

Task 09:

```
1 #!/bin/bash
2
3 dir="$1"
4 pattern="$2"
5
6 if [[ ! -d $dir ]]; then
7     echo "Directory not found!"
8     exit 1
9 fi
10
11 if [[ -z $pattern ]]; then
12     echo "Please provide a new file name pattern."
13     exit 1
14 fi
15
16 count=1
17 for file in "$dir"/*; do
18     if [[ -f $file ]]; then
19         ext="${file##*}"
20         newname="${pattern}${count}.${ext}"
21         mv "$file" "$dir/$newname" && echo "Renamed $file -> $newname" || echo "Failed to rename
22         $file"
23         ((count++))
24     fi
25 done
```

Output:

```
yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh
Directory not found!
yashraj@DESKTOP-A4HGUGP:~/Os$ |
```

Task 10:

```
1 #!/bin/bash
2
3 dir="$1"
4 days="$2"
5
6 if [[ ! -d $dir ]]; then
7     echo "Directory not found!"
8     exit 1
9 fi
10
11 if [[ -z $days ]]; then
12     days=30 # default: delete files older than 30 days
13 fi
14
15 # Delete old files
16 old_files=$(find "$dir" -type f -mtime +$days)
17 file_count=$(echo "$old_files" | wc -l)
18 echo "$old_files" | xargs -r rm
19 echo "Deleted $file_count files older than $days days"
20
21 # Remove empty directories
22 empty_dirs=$(find "$dir" -type d -empty)
23 dir_count=$(echo "$empty_dirs" | wc -l)
24 echo "$empty_dirs" | xargs -r rmdir
25 echo "Deleted $dir_count empty directories"
```

Output:

```
yashraj@DESKTOP-A4HGUGP:~/Os$ ./T6.sh
Directory not found!
yashraj@DESKTOP-A4HGUGP:~/Os$ |
```